

R01 Annex T — How Concrete Agent Technologies Act on the Trilemma

Oct 4, 2026 · @Ivan

1. Purpose, scope and status

On the documentation read, the three frameworks act on the risk arm through pre-effect barriers and on the cost arm through constants, but none supplies what sets the tier: a verification interface that returns many bits of binding information per unit cost. That interface belongs to the environment, not to the framework. This annex works that conclusion out, mechanism by mechanism, and lists what must be measured to confirm or refute it.

What this annex is. An annex to *R01 Viability as a Trilemma: Proof Sketches and Review Plan*. It reuses that document’s definitions, Lemma 1 and the tier table, and applies them to the three configurations that task T01 asks to qualify: Hugging Face smolagents, LangGraph (with LangSmith for tracing) and the OpenAI Agents SDK for Python. Each is read in its official documentation, opened on 4 October 2026.

What it is not.

- It is not an execution. Nothing here was run, installed or measured, and versions are not pinned.
- It is not a comparison of quality or safety between the frameworks, and it makes no claim about EA.
- It is not an admission decision. T02, T07 and T11 decide admission on evidence gathered against pinned versions.
- It does not cover the other technologies profiled elsewhere in the corpus (Microsoft Agent 365, Claude Agent SDK, FIWARE and others).

Evidence labels. Every row about a technology carries one label, so a reviewer can see at a glance what rests on documentation and what rests on the author.

Label	Meaning
D	Stated in an official documentation page opened for this annex.

Label	Meaning
I	Inferred by the author from documented behavior; needs a test before it is relied on.
U	Unknown from the pages read; a named test must settle it.

Documentation versions seen. The smolagents pages show release v1.26.0 in their version list and describe the API as experimental and subject to change. The LangGraph pages mention features that need `langgraph>=1.2.12` (interrupt response schemas) and `langgraph>=1.2` (run control). The OpenAI Agents SDK pages carry no release number on the pages read. T01 must pin exact releases, models, prompts and settings before any test.

Second reading. A second pass on 4 October 2026 closed most of the U labels in sections 4 to 6 and surfaced new hazards. Everything below is in the tables of those sections.

- **smolagents step callbacks are not a barrier.** The project blog states that step callbacks run at the end of each step, after the action has executed. The wrapper around the tools is therefore the only pre-effect barrier found (D).
- **LangGraph’s exit durability can lose progress.** In exit mode state is persisted only when execution exits or at an interrupt. A crash mid-run then restarts from the last exit, and effects since then repeat unless they sit inside tasks (D, I).
- **LangGraph’s RunControl is a cooperative stop.** It drains at a superstep boundary, so a deadline can be enforced between steps and not inside a long node. `recursion_limit` counts supersteps, not cost (D).
- **OpenAI sandbox agents allow rollback of confined effects only.** Snapshots restore the workspace, but they skip mounted remote storage, and the manifest describes only a fresh session, so live state can differ from it. That gap is a staleness hazard of the kind in section 9, and the receipts must record the effective state, not the manifest (D, I).
- **OpenAI’s `max_turns` counts model turns.** Its numeric default was not visible in the excerpts read.

2. Method: reading a mechanism through the trilemma

A mechanism matters to the trilemma only through one of five channels, so the method asks of each documented mechanism which channel it uses, where it sits in the review, decide, execute gate, and how its cost grows with the number of bits it must settle.

The three arms in technology-observable terms.

Arm	Formal quantity	What a technology run can observe	Unit
Cost	$R = c_e + gL + s$, where s is information spend	Model requests, input and output tokens, tool calls, wall time, human minutes, sandbox seconds	A vector, normalized as the adapter contract requires (E5); never silently collapsed to tokens
Risk	$\rho = P(V)$	Effects that violate the binding, read from environment-owned receipts, never from the framework's own trace	Probability per run
Efficacy	$\eta = P(T)$	A complete route delivered at technical quality of at least $J^* - \epsilon$	Probability per run

The five channels.

Channel	What the mechanism does	Arm it moves	What it cannot do
1. Barrier	Blocks or defers an effect before it happens	Risk	Create the information the barrier decides with; it relocates the information cost, it does not remove it
2. Information channel	Returns bits about the binding	Cost per bit and capacity per call, which set the tier	Return more bits than the environment holds

Channel	What the mechanism does	Arm it moves	What it cannot do
3. Replay and accounting	Decides whether work and effects are repeated, lost or double-counted on pause, resume or failure	Cost, and risk through duplicated effects	Make an irreversible violation reversible
4. Scale-only	Changes constants: cheaper calls, batching, parallelism, caching, smaller models	The threshold slack λ by a constant factor	Change the scaling class
5. Observation	Records what happened: traces, callbacks, logs	None directly; it decides whether the arms can be measured at all	Substitute for environment-owned receipts (E2, E5)

Position in the gate. The R01 contract fixes the order review, decide, execute, with a receipt visible only after the effect. Every mechanism is placed in one of four positions: before the decision, between decision and execution, at execution, or after the receipt. A barrier is only a barrier in the second position. The same check placed after the receipt is an observation.

The tier question. The tier is set by $f(b)$, the cost of settling b independent bits. A verification interface has two properties that matter: bits returned per call, and cost per call. Three interface shapes recur.

Interface shape	Bits per call	Typical cost growth	Tier
Per-bit probe: a membership test on one node or one route	At most 1	Linear in b	3
Map certificate: one call returns the whole binding map	L	Constant c_0 , plus transfer	1, subject to staleness (section 9)

Interface shape	Bits per call	Typical cost growth	Tier
Folded into the gate: the review that precedes execution already certifies the binding	L	Zero marginal	0

A deterministic code check and a model-based check can have the same capacity and very different cost per call. The capacity decides the tier and the cost per call decides the constant.

Two attribution rules. They exist to stop a mechanism being credited with something it cannot do.

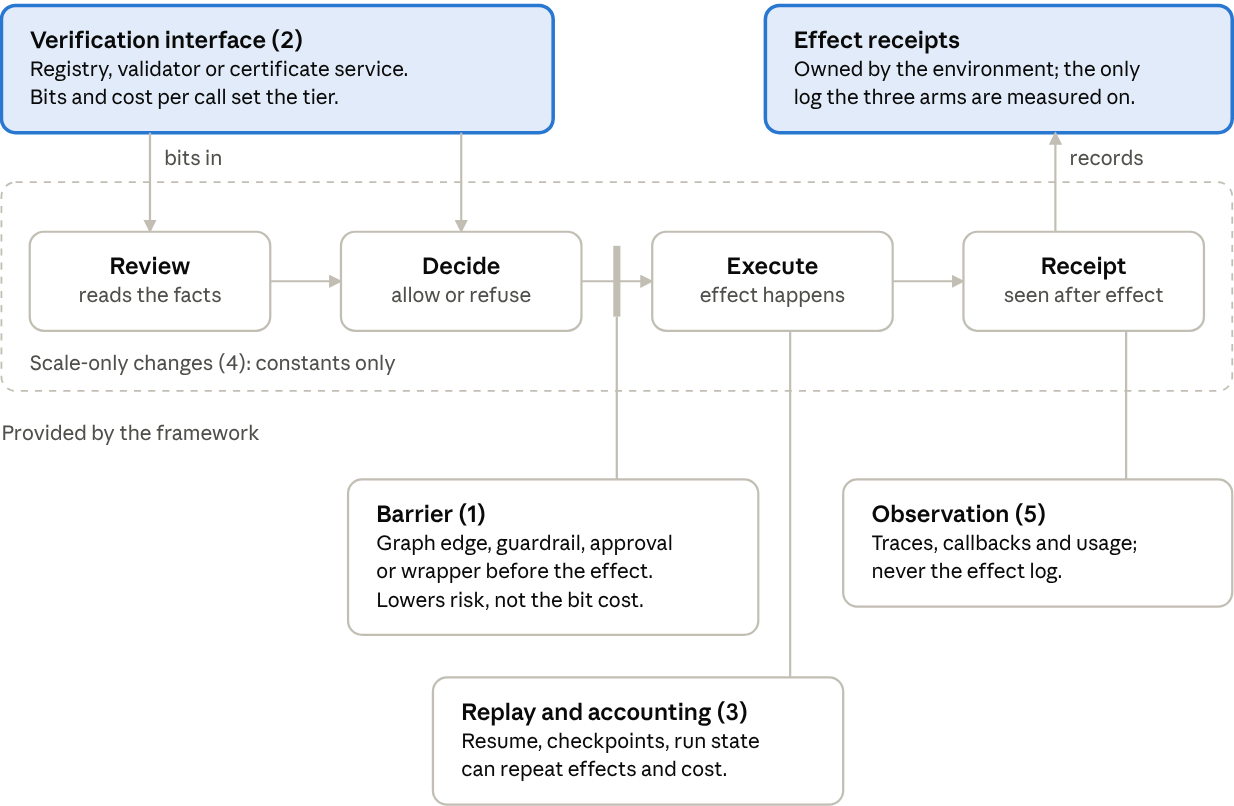
- 1. **Source rule.** A mechanism that decides using information must name the source of that information. If the source is the model's own estimate of the binding, it adds no information. By Lemma 1 it can only reshuffle risk and add cost. A guardrail model that does not read the environment's binding facts gives the same prior as the agent.
- 2. **Coverage rule.** A barrier credited with lowering risk must be shown to act before the effect on every effectful tool in the stack. One unguarded effect path voids the claim for that path.

3. The measurement sheet

Sixteen fields, filled in for each configuration at T01 and T02, are enough to place any mechanism in a channel, a gate position and a tier. The drawing shows what owns what and where each channel attaches, and the sheet lists what must be extracted so that no cell of the tables in sections 4 to 6 rests on memory.

Frameworks host the gate; the environment supplies the bits

Owned by the environment



Provided by the framework

gate and channels · who owns what

The environment’s verification interface sets the tier. The barrier, replay and observation mechanisms a framework provides change risk, duplicated effects and measurability, and the effect receipts stay with the environment.

The sheet.

#	Field	What to extract	Where it enters
1	Pinned versions	Release, commit, model identifier, prompts and settings	Every arm; transfer evidence E4
2	Effect surface	Every tool that can cause an effect, with its type: function tool, local MCP tool, hosted tool, built-in execution tool, code action	Coverage rule; X2

#	Field	What to extract	Where it enters
3	Effects per decision	Effects executed per model decision, including batched and parallel ones	Unit g; receipts between effects; X6
4	Pre-effect hooks	Every hook that can run before an effect, and whether it can veto it	Barrier channel; X1, X2, X7
5	Hook mode	Blocking or parallel, synchronous or asynchronous	X1
6	Cost of a check	Model requests, tokens, compute, latency and human minutes per check, kept as separate dimensions	f(b), a, c; E5
7	Bits per check	The size of the output alphabet of each verification call	Capacity per call, which sets the tier
8	Information source	What each check reads: a registry, the model's own estimate, a human's knowledge	Source rule
9	Approval semantics	Scope (call, tool identity, run), persistence, and behavior on malformed input	Sticky-decision hazard; X5
10	Replay behavior	What re-runs on resume, for each durability or resume mode	Invariant 7; X3
11	Native cap	Unit and default of the step, turn or recursion cap, and any token or cost cap	Cost arm
12	Native usage accounting	Whether usage includes guardrail, sub-agent and compaction calls	E5; X8
13	Native record	What is recorded, how it is exported, and how it fails	Invariant 8; X9
14	Confinement	Where effects execute, whether they can be discarded, and whether a commit step exists	Assumption A4; X13

#	Field	What to extract	Where it enters
15	Multi-agent scope	Whether barriers and guardrails cover handoffs, managed agents and agents used as tools	Coverage rule
16	Reversibility of each effect tool	Per effect tool, whether the effect can be undone	Assumption A4

4. smolagents

On the pages read, smolagents has no framework-level pre-effect gate in its core loop, so the barrier has to be an environment-owned wrapper around the tools. Its distinctive effect on the trilemma is batching: code actions and parallel tool calls let several effects occur between model decisions, which lowers cost and removes the intermediate receipts that a gate relies on.

Profile (D). smolagents is a library for multi-step agents that follow a ReAct loop of thought, tool call, observation. It offers two agent types: `CodeAgent`, which writes its tool calls as Python code and is the default, and `ToolCallingAgent`, which writes them as JSON. The reference describes the API as experimental. The R01 plan asks T01 to assess `ToolCallingAgent` first and to treat `CodeAgent` as a distinct profile if its execution surface is needed.

Mechanisms and their channel.

Mechanism	Evidence	Position in the gate	Channel	Effect on the arms	Moves the tier?	Label
ToolCallingAgent: JSON tool calls through the model's tool-calling interface	Agents reference	At execution	Baseline	One model request per decision step sets the unit g	No	D
CodeAgent: actions as Python code, with composition of several tool calls and logic in one step	Secure code execution, Agents reference	At execution	Scale-only, with a barrier-coverage side effect	Fewer model requests per effect, so g falls; no receipt between effects inside one block	No	D, I

Mechanism	Evidence	Position in the gate	Channel	Effect on the arms	Moves the tier?	Label
max_tool_threads: parallel tool calls in ToolCallingAgent	Agents reference	At execution	Scale-only	Concurrent effects with no intermediate receipt; order not fixed	No	D, I
max_steps, default 20	Agents reference	Whole run	Cost cap	A hard cap R in steps. The pages read document no cap in tokens or currency	No, but it makes the band explicit	D
planning_interval	Agents reference	Before the decision	Scale-only, efficacy	Extra planning model calls at the chosen interval; may raise eta at a cost	No	D, I
step_callbacks, a list or a dict keyed by step type	Agents reference, plan customization example, smolagents project blog	At the end of a step, after the effect	Observation, and a human gate where documented	Callbacks run at the end of each step, after the action has executed and been logged to memory. The documented use pauses after a planning step, and the project blog shows a callback editing memory. A step callback therefore cannot veto the effect of its own step	No, for the effect of its own step	D (documentation and project blog); confirm at X7

Mechanism	Evidence	Position in the gate	Channel	Effect on the arms	Moves the tier?	Label
Human plan review: approve, modify or cancel after the plan is created	Plan customization example	Before the decision	Information channel (a human)	A human can supply plan-level information, at a cost in human minutes and bounded by what the reviewer knows	No, unless the reviewer reads a binding map	D, I
final_answer_checks: validation functions run before a final answer is accepted	Agents reference	After execution	Observation, efficacy gate	Can reject an incomplete answer, so it acts on eta. Effects have already happened, so rho is unaffected	No	D, I
LocalPythonExecutor: AST interpreter, imports disallowed unless authorized, cap on operations	Secure code execution	At execution	Barrier for code-level escapes only	Protects the host. It does not know the binding, and effects through authorized tools are untouched	No	D
Remote executors: Blaxel, E2B, Modal, Docker, Pyodide with Deno	Secure code execution, project README	At execution	Barrier on host blast radius	Changes the irreversible-violation assumption only if the effect tools themselves run inside the sandbox and their effects can be discarded	Possibly (A4)	D, I

Mechanism	Evidence	Position in the gate	Channel	Effect on the arms	Moves the tier?	Label
Sandbox scope: run snippets only, or run the whole system	Secure code execution	Design choice	Constraint	Snippet-only mode does not support managed agents and still passes state; whole-system mode supports them but may need secrets inside the sandbox	No	D
managed_agents, a multi-agent hierarchy	Agents reference	Decision and execution	Efficacy, cost	Delegation adds model calls. Whether sub-agent effects are gated by the manager is not stated	No	D, U
Memory and resume: run with reset set to false keeps prior steps; replay prints steps	Plan customization example, Agents reference	After the receipt	Replay and accounting	Resume keeps memory. The reference describes replay as a printed replay of steps, not re-execution	No	D
interrupt(): interrupts the agent execution	Agents reference	Not stated	Possible barrier	The pages read do not say at which point in a step the interruption takes effect	Unknown	U
Telemetry tutorial (inspect runs)	Navigation only	After the receipt	Observation	Listed but not read; T02 must read it	Unknown	U

Where the barrier lives. Tools are plain callables (I), so the natural pre-effect barrier is a wrapper owned by the environment, which is what the R01 adapter contract already asks for under the name controlled tool wrappers. That wrapper is also where binding information enters, through a call to whatever registry the environment exposes. The tier of a smolagents configuration is therefore the tier of the wrapper's verification interface. smolagents itself neither raises nor lowers it.

Batching and the slack threshold. Let m be the number of effects executed per model decision. With m above 1 the policy cannot read a receipt between effects, so inside a block it behaves like the blind policy. At the same time the model-request share of the gate falls from about g to about g/m , which lowers g . Because the threshold slack is $\lambda^* = c(1 - \alpha)/g$, a smaller g raises λ^* . The consequence is counter-intuitive and testable: cheaper execution widens the relative trilemma band, because information becomes relatively more expensive.

Sandboxing and the irreversible-violation assumption. Assumption A4 says a violation is permanent. A sandbox changes that only if the effect itself lands inside the sandbox and is committed to the outside world by a separate step the environment can veto. A route-level dry run of this kind returns per-node activity for the whole route, so its capacity is L bits per call. Its cost, however, grows with route length, because executing L nodes in the sandbox costs about L times the dry-run cost per node. That is Tier 3 with a small constant, not Tier 1. If the effect tools stay outside the sandbox, nothing changes.

Plan review and final-answer checks. A human reviewing a plan supplies information only if the reviewer knows the binding, and each review costs minutes. The documented interface of three choices carries at most $\log_2(3)$ bits per review, although modification of the plan can carry more if the reviewer knows more. A final-answer check cannot lower risk because it runs after the effects. Rejection of an incomplete answer also consumes steps from `max_steps`, so it trades cost for efficacy.

What T02 and T07 must settle for smolagents.

1. Confirm with X7 that a step callback cannot veto the tool call of its own step, as the project blog states that callbacks run at the end of each step (D).
2. Effects executed per decision step for `CodeAgent` and for `ToolCallingAgent` with parallel threads, counted from environment receipts (D, I).
3. Where tools execute under each remote executor, and whether their effects can be discarded (I).
4. What `interrupt()` does and when (U).
5. What the telemetry tutorial exposes and whether it is complete (U).

5. LangGraph and LangSmith

LangGraph is the configuration whose structure maps most directly onto the R01 gate, because review, decision and execution can be separate nodes with the decision as an edge. Its main effect on the trilemma is in replay: pauses and resumes re-run code, so the

same machinery that gives a barrier can duplicate effects if the effect sits on the wrong side of a pause.

Profile (D). LangGraph is a low-level orchestration framework and runtime for long-running, stateful agents. It lets a graph mix deterministic, hand-coded steps with model-driven steps. Persistence, durable execution, streaming and human-in-the-loop are its stated core capabilities. LangSmith is the separate platform for tracing and evaluation. The R01 plan asks T01 to check checkpoint and replay semantics, duplicate effects, observation isolation and accounting for resumed work, and states that persistence grants no authority and certifies no evidence by itself.

Mechanisms and their channel.

Mechanism	Evidence	Position in the gate	Channel	Effect on the arms	Moves the tier?	Label
Graph of nodes, edges, conditional edges and Command with goto	Overview, interrupts page	All positions	Structure	Review, decide and execute can be separate nodes. If the execute node is reachable only through the decision edge, the decision is a barrier by construction	No	D, I
Deterministic steps mixed with model-driven steps in one graph	Overview	Before the decision	Scale-only	A review written as code costs compute, not tokens, so the cost per check can be far below a model-based check	No, it lowers c	D
Checkpoint and thread_id	Persistence, interrupts pages	After the receipt	Replay and accounting	Graph state is saved per step and loaded by thread_id on resume. Production needs a durable checkpoint, since the in-memory ones lose state on restart	No	D

Mechanism	Evidence	Position in the gate	Channel	Effect on the arms	Moves the tier?	Label
<code>interrupt()</code> with <code>Command(resume=...)</code>	Interrupts page	Before the effect when placed before it	Barrier, and an information channel from a human	Pauses until resume, with no time limit stated. The payload must be JSON-serializable. The resume value becomes the return value of the call	Human bits, bounded by what the reviewer knows	D
Typed response schema on interrupt (needs langgraph 1.2.12 or later)	Interrupts page	Before the effect	Information channel	A Pydantic, TypedDict or dataclass schema validates the resume value. A plain JSON Schema dictionary is exposed but not validated	No	D
Node restarts from its beginning on resume	Interrupts page	Replay	Replay and accounting	Code before the interrupt runs again. Side effects before it should be idempotent, or each resume repeats them	No, but raises cost and can repeat a violating effect	D
Index-based matching of several interrupts in one node	Interrupts page	Replay	Replay and accounting	Skipping or reordering an interrupt between runs makes resume values attach to the wrong interrupt	No	D for the rule, I for the consequence
Parallel interrupts resumed through an id-to-value map	Interrupts page	Before the effect	Information channel	Each pending interrupt must be answered with its own id	No	D
Static interrupts: <code>interrupt_before</code> and <code>interrupt_after</code>	Interrupts page	Before or after a named node	Barrier	Pauses around a node. The documentation presents them for debugging and does not recommend them for human-in-the-loop	No	D

Mechanism	Evidence	Position in the gate	Channel	Effect on the arms	Moves the tier?	Label
Durability modes: exit, async, sync	Durable execution	Replay	Replay and accounting	Higher durability adds overhead. In exit mode changes are persisted only when execution exits, by success, by error or at a human-in-the-loop interrupt. In async there is a small risk that checkpoints are not written if the process crashes	No	D
Tasks around side effects and non-deterministic operations	Durable execution	Replay	Replay and accounting	The documentation advises wrapping side effects in tasks and designing for determinism and idempotence. That completed work is not re-run on resume is the intended purpose	No	D for the advice, I for the mechanism
Time travel from checkpoints	Persistence page	Replay	Replay and accounting	Listed as a use of checkpoints. Re-running from an earlier checkpoint can re-run nodes with side effects	No	D for existence, I for the risk
Store for long-term memory across threads	Persistence page	Before the decision	Information reuse	Could hold a cached registry answer across threads. Reuse is free after first acquisition and can go stale	Possibly, through caching	D, I
Subgraphs called as functions	Interrupts page	Replay	Replay and accounting	The parent node and the subgraph node both restart from the beginning on resume	No	D

Mechanism	Evidence	Position in the gate	Channel	Effect on the arms	Moves the tier?	Label
LangSmith tracing and evaluation	Overview	After the receipt	Observation	Traces runs and evaluates outputs. Its Engine detects issues in traces and proposes fixes, which is detection after the fact	No	D
RunControl (needs langgraph 1.2 or later)	Durable execution snippet	At a superstep boundary	Cost and deadline cap (I)	Created and passed as control to a run. The excerpt describes drain state read through the runtime, to adjust node behavior before the superstep boundary is reached. It reads as a cooperative stop at a boundary, not inside a node. The hard cap in the documentation is <code>recursion_limit</code> , which counts supersteps, not cost	No	D for existence, I for use as a cap

Structure gives a native barrier; it does not give information. A graph can route every effect through a review node and an edge, and then the effect node cannot run unless the edge allows it. What the review node knows is whatever it can read. If it reads a registry that answers per node, the interface is a per-bit probe and the tier is 3 with a small constant. If it reads a map certificate in one call, the interface is Tier 1. LangGraph supports both and chooses neither.

Replay as a risk mechanism. Every approval is a resume, and every resume restarts the node. If a non-idempotent effect sits earlier in the same node than the interrupt, each resume repeats it. The effects then multiply with the number of approvals, each one costs g again, and a violating effect is repeated while the violation counter stays at one. The documentation describes the replay of a loop of interrupts as growing with each resume,

because the first resume replays one iteration, the second two, and so on. The design consequence is that the interrupt must come before the effect and the effect must sit in its own node.

Human approval is a one-bit probe behind a barrier. An approve-or-reject interrupt on a route is a membership test returning at most one bit per decision. Because rejection prevents the effect, it is risk-free. That is the Tier 3' shape of the first document, with c equal to the cost of one human decision, in minutes and in deadline. The wait has no stated limit, so it consumes the R01 deadline T as well as budget.

Durability modes and the recorder. In sync mode every checkpoint is written before execution continues, which costs time. In async mode a crash can lose a checkpoint, and after resume the node may run again although its effect already happened. A checkpoint is therefore not a receipt. The environment must own the append-only record of effects, as the R01 adapter contract requires, and the checkpointer can only be compared against it.

LangSmith. Tracing is an observation channel. The Engine, which detects issues in traces and proposes fixes, acts on later runs. It does not change the risk of the run in which an effect has already happened. Its relevance is to drift and to measurement, not to the arms of a single run.

What T02 and T07 must settle for LangGraph.

1. Count effects per route under resume in each durability mode, with and without tasks around the effect.
2. Test approval-to-interrupt matching when node code is non-deterministic.
3. Test whether a static interrupt placed before the effect node is a reliable barrier, and whether it survives replay.
4. Confirm that RunControl stops a run only at a superstep boundary, and read the time-travel semantics, which this annex knows only from a secondary source that describes replay as re-executing the nodes after a checkpoint.
5. Compare the cost of a graph step with the cost reported by LangSmith, so E5 accounting is complete.

6. OpenAI Agents SDK

The OpenAI Agents SDK has the richest set of documented pre-effect controls of the three, but each is conditional on configuration and tool type, and its default configuration is not a barrier. Its distinctive effect on the trilemma is that a rejection-based barrier turns

every blocked attempt into a paid probe, which quantifies how wide the trilemma band is when information is obtained by trying and being refused.

Profile (D). The Python SDK provides agents, tools, handoffs, guardrails, sessions, human-in-the-loop approvals, tracing and usage tracking. The R01 plan asks T01 to check whether trace hooks expose every commitment, attempt and effect, to add environment-owned receipts where they do not, and states that SDK traces alone do not prove permission or legitimate completion.

Mechanisms and their channel.

Mechanism	Evidence	Position in the gate	Channel	Effect on the arms	Moves the tier?	Label
Tool input guardrails on function tools and on local MCP tools whose server configures guardrails	Guardrails page	Between decision and execution	Barrier	Run before the tool executes; can skip the call, replace the output with a message, or raise a tripwire	No	D
Tool output guardrails	Guardrails page	After execution	Observation	Run after the tool has executed; can replace the output or raise a tripwire. The effect has already happened	No	D
Tools outside the guardrail pipeline	Guardrails page	At execution	Coverage gap	Hosted tools (web search, file search, hosted MCP, code interpreter, image generation) and built-in execution tools (computer, shell, apply patch, local shell) do not use it. Handoff calls are not covered, and Agent.as_tool exposes no tool-guardrail options directly	No	D
Agent-level input guardrails, parallel mode (the default)	Guardrails page	Overlaps with the decision and with execution	Not a barrier	The guardrail runs alongside the agent. The documentation states the agent may already have consumed tokens and executed tools before being cancelled	No	D

Mechanism	Evidence	Position in the gate	Channel	Effect on the arms	Moves the tier?	Label
Agent-level input guardrails, blocking mode	Guardrails page	Before the decision	Barrier on the first input only	The agent does not start if the tripwire fires. Runs only for the first agent in a chain	No	D
Output guardrails	Guardrails page	After execution	Observation	Run only for the agent that produces the final output. When a terminal tool result is the final output, the tool has already run	No	D
A guardrail implemented as a model call	Guardrails page example	Varies	Cost	The documented example runs a second agent as the check. Each check is then a model request. Whether those requests are counted in the main run's usage is not stated	No, it adds cost	D for the example, U for accounting
needs_approval on function tools, Agent.as_tool, shell and apply-patch tools; require_approval on MCP servers	Human-in-the-loop page	Between decision and execution	Barrier, and an information channel	Pauses a run until approved or rejected. Approval rules fail closed when arguments are missing, malformed or not an object	Human bits, or code bits through a callback	D
Programmatic approval callbacks: on_approval and on_approval_request	Human-in-the-loop page	Between decision and execution	Barrier with a code decision	Shell, apply-patch and hosted MCP tools can approve or reject in code without pausing for a person	A registry call at constant cost is possible here	D
Sticky decisions: always_approve and always_reject	Human-in-the-loop page	Replay	Certificate reuse	Persist a decision for the same tool identity for the rest of the run, and survive serialization of the run state	No, and a hazard (below)	D
pre_approval_tool_input_guardrails	Guardrails page	Before the approval pause	Barrier	Runs tool input guardrails before the approval interruption; they are checked again after approval	No	D

Mechanism	Evidence	Position in the gate	Channel	Effect on the arms	Moves the tier?	Label
RunState: serialize, store, resume	Human-in-the-loop page	Replay	Replay and accounting	Holds approvals, pending calls, arguments and usage. The page states that recovery must reconcile tool side effects before retrying, and that snapshots are not authenticated	No	D
Tracing, on by default	Tracing page	After the receipt	Observation	Records generations, tool calls, handoffs and guardrails. Unavailable under zero data retention. Exports in batches every few seconds, and a failed batch is dropped	No	D
Usage tracking	Usage page	After the receipt	Observation	Tracks requests and tokens per run, including compaction calls. Third-party adapters may not report usage by default	No	D
Turn cap	Guardrails page, Runner reference	Whole run	Cost cap	The Runner takes a <code>max_turns</code> argument with a default constant, and raises <code>MaxTurnsExceeded</code> when it is exceeded unless handled. The unit is the model turn. The numeric default was not visible in the excerpts read	No	D, U

Mechanism	Evidence	Position in the gate	Channel	Effect on the arms	Moves the tier?	Label
Sandbox agents with Docker and local sandbox clients, a manifest, permissions and snapshots	Sandbox agents guides, as search-result excerpts	At execution	Possible reversibility	Beta in the Python documentation. A persistent workspace the model can search, edit and run commands in. A manifest describes a fresh workspace, and a reused session or snapshot may be the effective source of state. Snapshots let a later run resume, and a lost container is restored from the last checkpoint. Snapshots skip mounted remote storage. Effects confined to the workspace can be undone by restoring a snapshot (I). Effects through tools on external systems cannot	Possibly (A4)	D, I

The barrier is conditional. Three configurations act before the effect: a tool input guardrail on a covered tool, a blocking input guardrail, and an approval requirement. The default parallel input guardrail does not, by the documentation's own statement. Any hosted or built-in execution tool bypasses the guardrail pipeline. The coverage rule of section 2 therefore applies directly: an R01 effect must be a function tool, or a local MCP tool with guardrails configured, otherwise the barrier claim is void for it. The R01 contract keeps native defenses enabled, so default and blocking modes must be registered as separate configurations. Otherwise the phrase native defenses names two different systems.

Where information enters. A guardrail function is ordinary code, so it can read a registry and decide deterministically at a cost of compute. It can also call a model. The source rule decides the value of either. A model guardrail that cannot read the environment's binding facts adds a model request and no information. A guardrail that reads the registry returns exactly what the registry's interface returns.

Rejection-based probing and the width of the band. When a barrier rejects a call, the agent learns one bit and must try again. Each rejected attempt costs a model request and a guardrail evaluation, call it a . A blind policy under a barrier tries x on a layer, is refused if the binding is y , and tries y . In the worst case it pays two attempts per layer, so the budget needed is about twice R_{exec} , with risk zero and efficacy one. In the notation of the first document this is Tier 3' with $f(b) = a \cdot b$. If a is of the order of one model request and g is of the same order, the threshold slack is:

$$\lambda^* = \frac{a}{(1-\alpha)g}; \approx 1-\alpha$$

The trilemma band then extends to budgets close to twice the execution budget. This is a derived consequence (I) resting on a being about g , and it is the first number that T05 and T06 should measure, because it says that a barrier alone does not buy cheap safety. A band much narrower than 1 would need a to be much smaller than a model request, which means checks that are not preceded by a model decision, such as a route-level validator in code.

Approvals and the sticky-decision hazard. `needs_approval` with a callback is a one-bit probe behind a barrier, in the same way as a `LangGraph` interrupt, with c equal to a human decision or a code decision. The sticky option is different in kind. A persisted decision for a tool identity approves later calls to the same tool whatever their arguments. In the R01 fixture the effect tool is the same function for every node, so one sticky approval would approve every later node whatever its binding. A certificate must be scoped to the node and to the binding version. A sticky approval is scoped to neither unless the tool identity itself encodes the node.

Recorder and resume. Traces are batched, can be dropped when export fails, are unavailable under zero data retention, and may omit inputs and outputs when sensitive-data capture is off. They are an observation of the run, not the record of effects. On resume, the documentation itself places on the application the duty to reconcile tool side effects before retrying, which is the same duplicate-effect problem as in `LangGraph`.

What T02 and T07 must settle for the OpenAI Agents SDK.

1. Whether an effect tool executes before the tripwire in default parallel mode, using a slow guardrail and an effect tool that writes an environment receipt (D, test).
2. The pipeline coverage of every effect tool in the fixture, by tool type.
3. Whether guardrail model calls appear in the main run's usage (U).
4. Whether a sticky approval carries across nodes with different bindings.

5. Trace completeness against environment receipts under normal export, failed export and disabled tracing.
6. Read the sandbox agents guides in full, and pin the numeric default of `max_turns` (U).
7. The ratio a/g for a rejection-based protocol, measured in model requests and in tokens.

7. Cross-technology comparison

The three frameworks differ more in where they place the barrier, how they replay and how they count cost than in anything that sets the tier, and the tier is set by the verification interface the environment exposes. The tables below make that comparison explicit. Every cell marked I or U is a hypothesis for T02 and T07, not a finding.

Levers by arm.

Arm	smolagents	LangGraph and LangSmith	OpenAI Agents SDK
Cost: unit of the hard cap	Steps (<code>max_steps</code> , default 20)	Supersteps (<code>recursion_limit</code>); the default depends on the version, so pin it (D)	Turns (<code>max_turns</code>); the numeric default was not visible in the excerpts read (D, U)
Cost: main constant levers	Batching through code actions and parallel tool threads; <code>planning_interval</code> adds calls	Deterministic nodes keep checks out of the model; durability mode adds overhead	Model-based guardrails add a request per check; blocking mode adds latency
Risk: pre-effect barrier	Wrapper around tools (I); step callbacks run after the step's effect (D)	Graph edge before the effect node, interrupt before the effect, static interrupts	Tool input guardrails, blocking input guardrail, <code>needs_approval</code>
Risk: known gaps	Batching removes receipts between effects; local executor protects the host only	Replay repeats code before an interrupt; async durability can lose checkpoints	Parallel default is not a barrier; hosted and built-in tools bypass guardrails; sticky approvals

Arm	smolagents	LangGraph and LangSmith	OpenAI Agents SDK
Efficacy levers	planning_interval, final_answer_checks, managed agents	Deterministic and model-driven steps in one graph; human edit of state	Handoffs, output guardrails, approvals with rejection messages
Native record	Agent memory and replay of steps (D)	Checkpoints, and LangSmith traces	Traces and usage, with export caveats
Record usable as the effect log	No, not an environment receipt	No, a checkpoint is not a receipt	No, a trace is not a receipt

Which interface shape each framework can host.

Interface shape	smolagents	LangGraph	OpenAI Agents SDK
Barrier before the effect	Environment wrapper (I)	Native through structure (D, I)	Native for covered tools (D)
Human approval before the effect	Plan level documented; step level U	interrupt (D)	needs_approval (D)
Code decision at constant cost	Inside the wrapper (I)	Deterministic node (D)	Guardrail function and approval callbacks (D)
One-call map certificate	A tool or wrapper could read it (I)	A node could read it (I)	A guardrail or tool could read it (I)
Route-level dry run in a sandbox	Executors for code (D, I); effect tools unclear	Not found in the pages read (U)	Sandbox agents (U)
Verification folded into the gate	Not provided	Not provided	Not provided
Durable resume of a paused gate	Memory with reset false (D); effect semantics not stated	Checkpoint and thread_id (D)	RunState (D)

No framework supplies the registry, the map certificate or a gate that already certifies the binding. Each can host a client for one if the environment provides it. This is the table's main reading: the frameworks are hosts for interfaces, not sources of binding information.

Prior tiers by stack, under plausible environment interfaces. These are expectations to be replaced by measurement.

Stack	Verification interface	f(b)	Tier	Expected band	Note
Model decides alone, no barrier, any framework	None	Not applicable	Blind policy	Cost and efficacy attainable together; risk high	The worst corner of the trilemma
Barrier with rejection, registry answers per node	One bit per refused attempt	a b, with a about one model request	3'	Slack up to about (1 - alpha)	Risk-free; cost near twice execution in the worst case
Human approval per node or per route	One bit per decision	Human minutes per bit	3'	Set by human cost, plus deadline	Wait has no stated limit in LangGraph
Deterministic route validator in code before any effect	The violations of a whole route in one call	Constant c0	1	Width about c0	Valid only while the binding does not drift
Map certificate fetched once and cached in state or store	All L bits in one call	Constant c0, then free	1 until stale	Width about c0	Staleness reopens a per-effect cost (section 9)
Route-level dry run in a sandbox	The activity of every node of the route	About c_dry per node	3 with small c	Width about c_dry times L	Only if effects can be discarded
Review inside the gate, guaranteed by the environment	All L bits	Zero marginal	0	None	A change of contract, not of framework

What this implies for T11, as a prior and not a ranking. For mapping effort on E1 and E2, LangGraph's explicit nodes and edges suggest the smallest gap (I). The OpenAI SDK has the most native barrier and approval primitives, with coverage and mode conditions that must be registered. smolagents is the thinnest layer, so its barrier is wholly the adapter's,

and its `CodeAgent` batching is the largest hazard for the gate. T11 must replace this prior with observed accessibility, controllability and estimated cost, and it must retain all three candidates.

8. What no technology in this set changes

Ten invariants hold across smolagents, LangGraph and the OpenAI Agents SDK as documented, and each comes with the observation that would falsify it. They follow from Lemma 1 and from the R01 contract, and they are what a reader should keep if the details of any one framework change.

#	Invariant	Reason	Falsified by	Test
1	Information about the binding comes from the environment. A framework relocates when and where it is queried and what it costs, and it cannot create it.	Lemma 1 counts transcripts. A framework adds no transcript the environment did not produce	A framework component that reports binding facts the environment never exposed, verified against receipts	T02, T07
2	The tier is the growth of $f(b)$ for the verification interface in use, not a property of the framework.	The interface shapes of section 2 fix capacity per call	The same interface giving different growth under different frameworks beyond a constant factor	T05, T06, measurement protocol of section 10

#	Invariant	Reason	Falsified by	Test
3	A barrier lowers risk but not information cost. It turns the trilemma into a cost-efficacy dilemma in which the band has width about $a(1 - \alpha)L$, with a the cost of a refused attempt.	A refusal returns one bit and requires a new attempt	A barrier whose refusals return more than one bit at the same cost	T05, ratio a/g
4	A model-based check with no access to the environment's facts adds cost and no information.	The source rule	A model check that beats the prior on held-out bindings with no registry access	T04, T05
5	Human approval is a one-bit probe with a deadline cost, linear in the number of decisions.	Approve or reject returns at most one bit. LangGraph states interrupts wait indefinitely	A reviewer who reads a map and settles many bits in one decision, which moves the stack to Tier 1 at the cost of the map	T05
6	A violation stays irreversible unless the effect is confined and a separate commit step is under the environment's control.	Assumption A4	Effect tools that provably run inside a discardable sandbox, with commit under the environment	T03, T07

#	Invariant	Reason	Falsified by	Test
7	Pause and resume mechanisms duplicate work, and can repeat a violating effect, unless the effect is idempotent or isolated.	LangGraph documents node restart. The OpenAI SDK documents that recovery must reconcile side effects	A resume count of effects equal to one under every durability mode and approval pattern	T03, T04
8	No native trace or checkpoint is the effect log. The arms are measurable only against environment-owned receipts.	The R01 adapter contract. Traces are batched, droppable and can be disabled	A framework record that matches environment receipts exactly in all failure conditions	T03, T07
9	Constant-factor improvements never remove the band; they change the threshold slack λ^* . Cheaper execution can widen it.	$\lambda^* = c(1 - \alpha)/g$, so a smaller g raises it	A technology change that alters the growth class and not only the constants	T06, T12
10	A cached certificate or approval is valid only for its binding version. Under drift, per-effect revalidation restores linear cost.	Semantic stale-action behavior, discussed in section 9	A certificate that stays valid across a binding change, or drift too slow to matter in the tested horizon	T05, T12

Invariants 1 and 2 carry the main claim. If either is falsified, the classification in section 7 changes, and section 10 is built to test them first.

9. Composition, units and staleness

A stack of mechanisms reaches Tier 1 only if one component supplies a one-call map certificate and the stack keeps it fresh, and staleness turns the saving back into a per-effect cost whenever the binding can change between the certificate and the effect. Two further results sit beside it: the tier is stated per cost dimension, and one code-acting configuration can move probing from tokens to compute.

Stacks are not additive. Each component acts on one channel of section 2. A composite stack pays the sum of its components' costs, and its tier is the tier of the interface that actually delivers the bits. A barrier, a replay control and a recorder add constants and safety. They do not change the growth of $f(b)$.

Cost is a vector, and the tier is per dimension. The R01 contract asks for normalized resources and forbids silent collapse to one unit. A technology can move bits from an expensive dimension to a cheap one without changing the tier in the cheap one.

- In a code-acting agent with an environment wrapper that raises an exception when the barrier refuses, the probing loop can run in code. The model writes the loop once, inside one decision step, and each refused attempt costs a tool call in compute and time, not a model request (I).
- Information is then still a per-bit probe, so $f(b)$ is linear in compute and time. In tokens it is nearly flat, because the model wrote one loop.
- The trilemma band therefore persists in the dimensions where probing stays linear: compute, latency, registry rate limits and quotas. Whether it matters depends on the weights of the normalization. This is a testable effect of `CodeAgent`, and it is the strongest case in this annex of a technology helping without removing the trilemma.

Staleness: the cost of keeping a certificate valid (sketch). Take the map certificate of Tier 1. Suppose that between one effect step and the next, the binding of an upcoming layer changes with probability p , independently. If the certificate is refreshed every $k + 1$ effects, then to first order in p the expected number of stale effects is $pL(k + 1)/2$. Keeping the risk ceiling r then requires:

$$\begin{aligned} \mathbb{E}[\text{stale effects}] &\approx \frac{pL(k+1)}{2}; \\ &\quad \Longleftrightarrow \\ \#\text{refreshes} &\approx \frac{L}{k+1}; \\ &\approx \min\left(L, \frac{pL^2}{2r}\right) \end{aligned}$$

Three readings follow. If the total drift expected over the route is below the ceiling, that is pL of at most about r , one certificate suffices and the tier is 1. If it is above, the number of refreshes grows quadratically in L until it saturates at one per effect, which is the per-bit linear cost of Tier 3 with c equal to a refresh. Between the two, the stack is Tier 1 only in a regime that depends on the drift rate. The corpus's scenario 00I, in which a delayed action executes after a later repair has changed its decision basis, is the known case of this hazard. The sketch ignores the window between a check and its effect, which is the same hazard at a smaller scale and must be measured.

Stacks to test.

Stack	Components	Information cost	Tier	Valid while
S-A	Environment route validator and map certificate; framework gate node, edge or tool guardrail; environment receipts	One certificate plus refreshes	1, within the staleness regime	pL is of at most about r
S-B	Barrier with rejection; registry membership per node; the model re-plans after each refusal	a per bit, with a about one model request	3'	Always
S-C	Code-acting agent; environment wrapper that raises on refusal; probing loop in code	About a_{comp} per bit in compute and time; one model request for many attempts	3' in compute, nearly flat in tokens	Rate limits and latency budget hold; the model writes a correct loop
S-D	Human approval per route or per node	One human decision per bit	3'	The deadline T holds
S-E	No barrier and no registry	None	Blind policy	Not applicable

S-A, S-B and S-E are candidate comparator arms. Mapping them onto the benchmark arms of the corpus is left to T11 and to the benchmark owners. This annex takes no position on whether any Ecosystem Awareness component supplies the map-level interface of S-A. That is the differential the main R01 programme tests, and the annex gives no basis for a claim either way.

10. Experiments that would settle each unknown

Fourteen experiments settle the unknowns in sections 4 to 6 and test invariants 1 to 10, and the first nine need no model behavior at all. They test mechanism facts, such as whether an effect happens before a tripwire, and they can run with a scripted model, which the R01 plan allows for adapter plumbing only. The last five measure cost growth, drift, confinement and human cost. X13 can use a scripted model and X14 a simulated reviewer, and X10 to X12 need model-driven runs under their own registration.

Mechanism tests (scripted model; effect tools write environment-owned receipts).

ID	Question	Method	Measure	Reading of the result	Task
X1	In the OpenAI SDK default parallel mode, does an effect occur before a guardrail tripwire?	An effect tool that writes a receipt; a guardrail with a controlled delay; compare default parallel mode with blocking mode over delays from below to above model latency	Share of runs where the effect receipt precedes the tripwire	If positive, the default is not a barrier and the two modes are separate configurations	T04, T07
X2	Which effect tool types are covered by the guardrail pipeline?	One effect tool of each type: function tool, local MCP tool with guardrails, hosted tool, built-in execution tool, handoff, agent as tool	A matrix of covered or not covered, by environment receipt	Fixes which tool types may carry effects in the fixture (coverage rule)	T02, T07
X3	Does resume in LangGraph duplicate effects?	Effect before an interrupt in the same node, and in a separate node; durability exit, async and sync; crash injected before the checkpoint write; with and without tasks	Effect receipts per route	Settles invariant 7 for LangGraph and the safe placement of the effect	T03, T04

ID	Question	Method	Measure	Reading of the result	Task
X4	Does approval matching fail under non-deterministic node code?	A node that sometimes skips an interrupt; several interrupts per node; parallel interrupts	Resume values bound to the wrong interrupt	Tests the index-matching hazard	T04, T07
X5	Does a sticky approval carry across nodes with different bindings?	always_approve on the effect tool, then nodes whose binding is inactive	Violating effects approved	Tests the sticky-decision hazard	T04, T07
X6	How many effects occur per decision step in smolagents?	CodeAgent with a multi-call block and ToolCallingAgent with parallel threads against a scripted model	Effects per step, receipts between effects, model requests per effect	Measures batching and the unit g	T03, T05
X7	Can a smolagents step callback on an action step veto a call before execution?	Callback that raises or returns a refusal on an action step; effect tool with receipt	Whether the receipt exists	Confirms that the wrapper is the only barrier, or finds a documented exception	T02
X8	Are guardrail model calls counted in the OpenAI run's usage?	Count model requests on the provider or mock side and compare with the SDK usage object	Requests missing from usage	Completes E5 accounting	T03, T07
X9	Is any native record a complete effect log?	Compare SDK traces, LangGraph checkpoints and smolagents memory with environment receipts under normal operation, failed export, disabled tracing, redaction off and crash	Receipts missing from each native record	Settles invariant 8 and the recorder design	T03, T07

Cost-growth and drift tests (model-driven, registered under C11).

ID	Question	Method	Measure	Reading of the result	Task
X10	What is the cost of a refused attempt, and what is a/g ?	Stacks S-B and S-C on F_L with a barrier that refuses	Cost per refused attempt in model requests, tokens, compute and latency, and the ratio a/g with the predicted λ^*	Tests the claim that λ^* is of order one for rejection-based probing	T05, T06
X11	What is the tier of each stack on F_L?	Stacks S-A to S-E for L in 1, 2, 4, 8, 16, 32 at fixed cost per bit	The minimal budget $R^*(L)$ meeting the thresholds, per cost dimension	Classifies growth as constant, logarithmic or linear within the range	T05, T06, T12
X12	How does staleness behave under drift?	Stack S-A with an injected binding change at rate p between certificate and effect; refresh every $k + 1$ effects	Violating effects and refresh count against the sketch pL squared over $2r$	Tests the staleness result and invariant 10	T12
X13	Can effects be confined and discarded?	Effect tools inside and outside each remote executor; a commit step under the environment	Share of violating effects that can be discarded before commit	Tests assumption A4 and invariant 6	T07
X14	What does a human gate cost in deadline?	A simulated reviewer with a registered latency distribution; interrupts and approvals per node and per route	Decisions per route and time consumed against the deadline T	Measures c for Tier 3' with a human; no real reviewers without separate registration	T05

Tier-estimation protocol for X11. For each stack and each cost dimension, fix before any run the grid of L, the number of replicates, the seeds, the thresholds e , r and ϵ , the exclusion and retry rules and the models and policies, matched across frameworks where possible. Measure the overhead $R^*(L)$ minus $R_{\text{exec}}(L)$. Fit constant, logarithmic and linear models and classify by a pre-registered margin. A classification holds for the tested range only. With L up to 32 the experiment can separate a constant from a linear overhead when the per-bit cost is not negligible, and it cannot separate a logarithmic from a linear one. A stack that cannot be separated is reported as not distinguishable in the range, and no asymptotic claim is made.

Order. Run X2, X1, X7 and X9 first, since they decide which configurations exist and whether the record can be trusted. Then X3 to X6 and X8, which complete the adapter. Then X10 and X13 to X14, and last X11 and X12, which are the expensive ones and depend on all the others.

11. Threats to validity and what must not be concluded

The main threat is that this annex rests on documentation and on the author's inference, so its tables are hypotheses with a test attached, and the conclusions that survive a failed test are fewer than the ones stated. The threats below are ordered by how much of the argument each could overturn.

#	Threat	Why it matters	Mitigation
1	Documentation is not behavior	A page can be incomplete, wrong or out of date, and a mechanism absent from the pages read may exist in the product	Pin versions, run the mechanism tests X1 to X9, keep the D, I and U labels in every table
2	The APIs change	The smolagents reference calls its API experimental. LangGraph features are tied to release numbers	Record exact versions in every result and re-run after any upgrade
3	The family is synthetic	F_L has independent uniform bits, a fixed gate and a binding map the environment holds. Real bindings are correlated or structured, which lowers b^* (assumption A5)	State results as results about the family. Add correlated families before any claim about deployments
4	The environment-owned registry	The tier depends on an interface the environment may not have, or on facts that are ambiguous in real organizations	Treat the interface as an explicit parameter of each stack, and test several

#	Threat	Why it matters	Mitigation
5	Cost units	Cost is a vector. Normalization weights can reverse conclusions, as the code-acting case shows	Report every dimension separately and pre-register the normalization
6	Finite range against asymptotics	L up to 32 cannot separate logarithmic from linear growth	Report not distinguishable in the range and make no asymptotic claim
7	Model confounds	Differences between frameworks can be differences between models, prompts or settings	Match policies and models where possible, or declare the combined system being compared
8	Chains of inference	Derived figures, such as λ^* close to $1 - \alpha$ for rejection-based probing, rest on a being about g	Measure a/g directly in X10 before using the figure
9	Self-review	The annex was written and reviewed by one assistant	Independent review, as in section 12
10	Scripted-model runs	They test mechanisms and plumbing. They do not show autonomous behavior	Keep scripted and model-driven results in separate tables
11	Scope of the pages read	The smolagents telemetry tutorial was not read, and the sandbox agents guides and LangGraph RunControl were read only as search-result excerpts	Read them at T02 before any statement about them
12	Three frameworks only	Other technologies profiled in the corpus are outside this annex	Say so when quoting results, and extend with the same method

What must not be concluded.

- That any of the three frameworks is safer, cheaper or better than another. The annex ranks nothing.

- That the trilemma is an empirical fact about deployed agents. It is a sketch about a family, with measurements still to come.
- That a barrier solves risk. It does so only for covered tools in configurations that act before the effect.
- That any Ecosystem Awareness component gives an advantage. The annex takes no position.
- That a tier measured in a finite range holds beyond it.
- That a mechanism not found in the pages read does not exist.

12. Review plan, open items and sources

The review must try to break the classification, and it must be done by someone other than the author. Each step has a pass criterion that can fail.

Review plan.

Step	Object	Question	Method	Pass criterion	Task
1	Every D row in sections 4 to 6	Does the page say what the row says?	Reopen each cited page at the pinned version and check the claim against its section	Each D row traces to a page and section. Mismatches are corrected, not defended	T07
2	Every I row	Does the inference follow from the documented behavior?	Derive it again or refute it. List the rows that need a test	Each I row is derived, refuted or tied to an experiment	T07, T08
3	Sections 7 and 8	Is any invariant falsified by the experiments of section 10?	Run X1 to X9 first and update the tables	Labels move from D, I and U to measured results, and falsified invariants are rewritten, not removed	T04, T07
4	Formal consistency with the first document	Do λ^* , the Tier 3' width, the staleness sketch and the S-C unit argument follow from Lemma 1 and the tier table?	Re-derive each, check the independence and first-order assumptions	Each derivation is confirmed or narrowed	M03, M04, T08

Step	Object	Question	Method	Pass criterion	Task
5	Invariant 1	Does any framework component supply binding information the environment did not expose?	Adversarial search for such a component, and for a stack that folds verification into the gate	A component is found and the classification is restated, or none is found in the pages read and tests	T07
6	Whole annex	Does an independent reviewer reach the same classification?	Independent review of this annex and of the first document, with corpus coherence against E1 to E7 and the common criteria	Disagreements are logged with their evidence	T08, T10

Open items.

- Read in full the OpenAI sandbox agents guides and LangGraph's RunControl and time-travel pages, which were seen only as search-result excerpts, and the smolagents telemetry tutorial, which was not read.
- Pin the default hard-cap values: recursion_limit in LangGraph, which counts supersteps and whose default differs across versions, and max_turns in the OpenAI SDK, which counts turns and whose numeric default was not visible.
- Extend the method to the other technologies profiled in the corpus (Microsoft Agent 365, Claude Agent SDK and others).
- Add correlated and structured binding families, where b^* is below the number of layers.
- Study multi-agent configurations (managed agents, handoffs, agents as tools), where barrier coverage across agents is not guaranteed.
- Find what real permission systems return per query, to know whether a per-node, per-route or map-level interface is the realistic one.
- Turn the D, I and U labels into the machine-readable evidence matrix that T01 requires.

Sources opened for this annex.

Corpus, pinned at commit 419b8e9: [R01 README](#) and [Hugging Face remaining tasks T01 to T12](#).

smolagents: [Secure code execution](#), [Human-in-the-loop plan customization](#) and [Agents reference](#), release list showing v1.26.0. The mention of Pyodide with Deno comes from a search-result excerpt of the project README, which was not opened in full.

LangGraph: [Overview](#), [Interrupts](#) and [Persistence](#). The durability modes and the advice on tasks and idempotence come from search-result excerpts of the [Durable execution](#) page, which was not opened in full.

OpenAI Agents SDK: [Tracing](#), [Guardrails](#), [Human-in-the-loop](#) and [Usage](#). The sandbox agents, sessions, handoffs and running-agents pages are listed in the navigation and were not read.

Second reading, 4 October 2026, all as search-result excerpts and not opened in full: the smolagents project blog on step callbacks, the smolagents memory tutorial, the OpenAI sandbox agents guides, the OpenAI Runner reference, the LangGraph durable-execution page and the LangGraph recursion-limit how-to. A secondary source that summarizes LangGraph replay is used only to flag a test. The companion document is *R01 Viability as a Trilemma: Proof Sketches and Review Plan*.